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Task-01

1) Differnence between "Love and "Love" in NPL

→ In NPL, "Love" and "Love" are Treated as two different token because of case sensitivity. This can lead to duplication of feature and inconsistency in analysis. To avoid this, text is usually converted to lowercase during preprocessing so that both words are treated as the same token.

2) What happen if stopword are not removed?

→ If stopwords are not removed, that dataset my contain a lot of unnecessary and frequently occuring words like "is", "the", and "and". This increases noise, make the model less efficient, and can reduce accuracy because meaningful words get less importance.

3) Two real-world scenarios where removing stopwords can be harmful.

→ Removing stopwords can be harmful in sentiment analysis, where words like "not" change the meaning of a sentence (eg., "not good" vs "good")
It can also affect question-answering systems, where words like "what", "who", and "when" are

important for understanding the query.

4) Difference between stemming and lemmatization

→ Stemming is a simple technique that removes suffixes from words to reduce them to their base from, often without considering context (eg., "running" becomes "run"). Lemmatization is a more advanced method that uses vocabulary and grammer rules to convert words into their meanigful root form (eg, "better" becomes "good"). Lemmatization is more accurate but computationally expensive.